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A Note on the Initial Public Offering Process

As has been often noted in *Venture Capital and Private Equity*, the process of taking portfolio firms public is very important for entrepreneurial firms. While the claim of Black and Gilson that “a well developed stock market is critical to the existence of a vibrant venture capital market”¹ may be overstated, there is clearly a strong relationship. To be a successful entrepreneur, an understanding of the initial public offering (IPO) process is important.

This note summarizes the mechanisms by which firms go public. It highlights some of the key institutional features associated with these offerings and suggests some explanations for why the process works as it does. While we note differences across countries, our focus will be on the major industrialized country with the greatest volume of offerings, the United States. Although the note must of necessity summarize the complexity and details of these offerings, the references suggest some sources for further reading for those who wish to learn more about this often-mysterious process.²

Why Do Firms Go Public?

Firms and their investors typically have several motivations for going public. At the same time, some real costs may also be associated with such a transaction. The relative importance of these competing factors may vary across time and circumstances.

Potential Advantages One important motivation for going public is the need to raise capital. Many technology companies, such as new semiconductor manufacturers and biotechnology firms, require hundreds of millions of dollars to successfully introduce a new product. This kind of capital may be difficult to raise from other sources. Banks and other debt financiers, for instance, may consider the firm too risky to lend funds to. Meanwhile, even if a venture capital group was willing to finance such a company’s initial activities, it might not be able to continue funding the firm until it

¹ Bernard S. Black and Ronald J. Gilson, “Venture Capital and the Structure of Capital Markets: Banks versus Stock Markets,” *Journal of Financial Economics* 47 (1998): 243–277.

² This discussion is based in part on a variety of sources, especially Jay R. Ritter and Ivo Welch, “A Review of IPO Activity, Pricing, and Allocations,” *Journal of Finance* 57 (2002): 1795–1828; Josh Lerner, “ImmuLogic Pharmaceutical Corporation” (case series), Harvard Business School case Nos. 292-066 through 292-071, 1992; and Katrina Ellis, Roni Michaely, and Maureen O’Hara, “When the Underwriter is the Market Maker: An Examination of Trading in the IPO Aftermarket,” *Journal of Finance* 55 (2000): 1039–1074.

This note was prepared as the basis for class discussion.

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achieved positive cash flow. For instance, most private equity groups are restricted to investing no more than 10% or 15% of their capital in a single firm. Thus, the need to raise capital to finance projects may be an important motivation to go public.³

A second motivation is the desire to achieve liquidity. Entrepreneurs are likely to worry about placing “all their eggs in one basket” and will seek to achieve diversification by selling some of their shares. Private equity investors are also likely to desire to liquidate their investments in a timely manner, whether through outright sales of the shares or through the distribution of the shares to their investors, in order to achieve a high rate of return.⁴

Achieving liquidity, however, is typically not done at the time of the IPO. This is because of the fears of investment bankers, who worry that if insiders such as entrepreneurs and board members are seen as “bailing out” at the time of the offering, new investors will be unwilling to purchase shares. (Insider sales at the time of the IPO are more common among private equity-backed firms in Europe.) Thus, they seek to prohibit or severely limit the sale of shares at the time of offering and to restrict any additional sales during a “lock-up” period. (In addition, the speed and timing of sales by insiders may be restricted by government regulations, as is the case in the United States.) After the lock-up period expires, however, insider sales are likely.

A third motivation is that going public may help the firm in its interactions with customers or suppliers. Being a public firm can help a firm project an image of stability and dependability. This is particularly important in industries where products do not represent a one-time purchase, but require ongoing service or upgrades. For instance, a corporation may be unwilling to purchase software to run a critical function from a small private firm that might soon disappear and not be available to offer upgrades or address problems. Enhanced visibility is a particularly important rationale for foreign technology companies seeking to break into the U.S. market, who have increasingly chosen to go public on the NASDAQ exchange in New York rather than on their local exchange.

Potential Disadvantages At the same time, going public involves some real costs, which lead many firms to resist going public:

- The legal, accounting and investment banking fees from an offering are substantial, frequently totaling 10% of the total amount raised in the offering or more.
- The degree of disclosure and scrutiny associated with being a publicly traded concern may be troubling, especially for a family business that has been run as a private firm for several decades. In particular, since the enactment of the Sarbanes-Oxley Act of 2002, with its numerous requirement to prevent accounting fraud by firms listed on U.S. exchanges, an increasing share of new issuers have been listing at the lightly regulated Alternative Investment Market in London and elsewhere.⁵

³ It should be noted that many firms raise far more in follow-on offerings than they do in their IPOs. But the IPO may provide important advantages: even if the firm does not raise all the financing that it needs in the initial offering, it is likely to find a follow-on offering to raise more equity substantially quicker to arrange and less expensive after it is publicly traded.

⁴ For more about private equity distributions, see “Between a Rock and a Hard Place: Valuation and Distribution in Private Equity,” Harvard Business School case No. 803-161.

⁵ For a discussion, see http://www.capmktreg.org/pdfs/11.30Committee_Interim_ReportREV2.pdf.

- If a firm files to go public and the offering must be subsequently withdrawn, even due to factors beyond the company's control, some managers fear that the company may be "tainted." In particular, other investors may be reluctant to even consider investing in the concern, presuming that the reason that it was forced to withdraw its IPO was due to some ethical lapse or fundamental business problem.

Another complication is introduced by the fact that the market's appetite for new issues appears to vary dramatically over time. In particular, the volume of IPOs changes dramatically from year to year. These periods of high IPO activity appear to follow periods when stock prices have risen sharply. The bunching of offerings is even more dramatic when patterns are examined on an industry basis. During these periods, firms may find it significantly easier to sell shares in IPOs to investors.

What Is the IPO Process?

The process by which firms go public is a complex one. This summary highlights the crucial steps along this journey.

First Steps The first step in the going public process is the selection of the underwriter. Firms considering going public will frequently be courted by several investment banks. Among the criteria used by firms and their private equity investors to evaluate banks are the reputation of the research analyst covering the firm's industry, the commitments made to provide analyst coverage in the months or years after the offering, and the performance of past IPOs underwritten by the investment bank. One arena where investment banks very infrequently compete is in the pricing of the transactions. A fee of 7% of the capital raised, plus the legal and other costs borne by the bank, is standard across investment banks of both high and low caliber.⁶

In many cases, firms select multiple underwriters to manage the offering. These might include, for instance, a smaller investment bank that specializes in market segment or a particular industry and a larger bank with the ability to market equities very effectively to institutions, such as Goldman, Sachs or Morgan Stanley, termed a "bulge bracket" bracket firms in Wall Street parlance. Only one of the banks, however, will be designated as the lead, or book, underwriter. This firm will be responsible for the most critical function, the management of the records of who desires shares in the new offering and the allocation of the shares among investors. The managing or co-managing banks will in turn recruit other banks and brokerage houses to join the "syndicate," the consortium that will actually sell the offering to its clients. Thus, while only one to three banks will actually underwrite the offering, the number of financial institutions involved is actually much larger.

Even before the offering is marketed, the underwriter plays several important roles. These include undertaking due diligence on the company to insure that there are no "skeletons in the closet," determining the offering size, and preparing the marketing material. In collaboration with the law firm representing the firm, the investment bank will also assist in the preparation of regulatory filings.

In most major industrialized nations, permission from one or more regulatory bodies is required before a firm can go public. In the United States, these are the Securities and Exchange Commission (SEC) and state regulatory bodies. The review of the SEC focuses on whether the company has

⁶ In some small offerings, less prestigious underwriters may demand warrants from the firm in addition to a fee in cash. In some of the very largest offerings, the fee may fall as low as 5%. For a detailed discussion, see Hsuan-Chi Chen and Jay R. Ritter, "The Seven Percent Solution," *Journal of Finance* 55 (2000), 1105–1131.

disclosed all material information, not on whether the offering is priced appropriately. In past years, state regulators occasionally sought to assess whether an offering was fairly priced. (To cite one example, Massachusetts regulators had in December 1980 initially barred the sales of shares of Apple Computer in the state, even though it was an operating profitable company, on the grounds that its IPO price was too high.) Since 1996, however, all offerings being listed on one of the three major exchanges have been exempt from state-level scrutiny.

The extent of the disclosure required varies with the size of the offering and the firm. Many nations have provisions for simplified filings for smaller firms, or for those that will be listed on one of the smaller exchanges. In the United States, for instance, firms going public with less than \$25 million in revenues can use file Form SB-2 rather than the much more exhaustive S-1 statement, those raising less than \$5 million can file under Regulation A, which requires even less disclosure.

There may be other regulatory requirements as well. For instance, in the United States, the SEC designates the weeks before and after the offering as the “quiet period.” The firm’s ability to communicate with potential investors during this period (aside from the distribution of the offering document, also known as the prospectus, and formal investor presentations) is severely limited.

Marketing the Offering As the firm undergoes regulatory scrutiny, the investment bank begins the process of marketing the offering. It typically circulates a preliminary prospectus, or “red herring” (so named for the disclaimers typically printed in red on the document’s cover), to prospective institutional and individual investors in the firm. In many cases, the firm will also undertake a “road show,” in which the management team describes the company’s lines of business and prospects to potential investors.

The actual mechanism used to determine the price varies across countries. In the United States, “book-building” is the most frequently employed approach. In particular, the underwriter learns from potential investors how many shares will be demanded at each proposed price, which enables him to set the best price for the company. All indications of interest are recorded in a central “book” compiled by the lead underwriter. In many other countries, however, the share price is set before the information about demand is gathered (though a number of these countries, such as Great Britain and Japan, have recently adopted the U.S. system in hopes of stimulating IPO activity). Elsewhere, other systems are employed, such as formal auctions to determine the offering price. Auctions are also attracting attention in the U.S., given the enormous attention (though mixed success) of the Google IPO auction and the unquestioned success of the Morningstar offering.⁷

Reputable investment banks in the United States typically undertake only “firm commitment offerings.” In these transactions, unlike “best efforts” offerings, the investment bank commits to sell the shares to investors at a set price. This price, however, is not set until the night before the offering, so the actual risk that the investment bank runs of not being able to sell the shares is very small. This information gathered about demand proves invaluable during the “pricing meeting” on the night before the IPO. In this session, the investment bank and firm bring together all the information about demand in order to determine the price at which the shares will be sold to the public. In determining a price, the bankers are also likely to factor in information about valuation of comparable firms, as well as discounted cash-flow analyses of the firm’s projected cash flows.

The Day of the Offering and Beyond Whatever valuation is set at the time of the offering, the share price is likely to increase on the next trading day. (On average, even the first trade

⁷ See, for instance, Christine Hunt, “What Google Can’t Tell Us About Internet Auctions (And What It Can),” *University of Toledo Law Review*, forthcoming, http://papers.ssrn.com/sol3/papers.cfm?abstract_id=753625.

of the stock is at a substantial premium to the IPO price.) While the median firm undergoes only a very modest increase in its price, a small but significant number of firms have experienced a significant jump in their share price after going public. This was particularly true during the Internet bubble in the United States, where Internet companies such as Yahoo!, TheGlobe.com, and the Internet Capital Group, experienced jumps of several hundred percent on their first day of trading. But more generally, these types of high returns have been observed on the first day of trading across many nations and time periods.

Several explanations have been offered for this frequently observed pattern of high first-day returns:

- One possibility is that the increase in price (or the discount offered to investors who purchase IPO shares) is necessary to attract investors. Otherwise, uninformed investors might fear that they would be taken advantage of in offerings: for instance, informed investors would purchase most of the shares of promising firms, while leaving them holding the bulk of the unpromising offerings.
- A second possibility is that there is a “bandwagon” effect at work. Once sophisticated institutional investors indicate interest in a stock by buying shares, other less sophisticated investors “rush in” to purchase shares.
- A third explanation is that the investment bank frequently has “market power.” This view suggests that bankers deliberately set offering prices too low in order to transfer wealth to the select investors whom they let participate in the IPO. These investors, having reaped big returns on the first trading day, will presumably reward the bank by steering other transactions, such routine custodial services, to the bank. The litigation around the “spinning” of IPO shares to favored clients of major investment banks provided many dramatic illustrations of this phenomenon.

Each of these explanations is likely to capture some, but not all, of the complex phenomenon of IPO pricing.

Another commitment made by underwriters in the United States is to stabilize the price in the days and weeks after the offering. Typically, the underwriter will try to prevent the share price from falling below the offering price. In undertaking this stabilizing activity, the investment bank will almost always employ the “Green Shoe” option, a complex feature named after the 1963 offering where it was first employed. Essentially, investment bankers reserve the option to sell 15% more shares than the stated offering size. The investment banker will often sell 115% of the projected offering size: for instance, if the firm announced its intention to sell 2 million shares, the investment bank would actually sell 2.3 million. If the share price rises in the days after the offering, the bank simply declares the offering to have been 15% larger than the size projected initially. If the share price drops below the initial offering price, however, the bank will buy back the additional 15% of shares sold. This will allow the bank to help fulfill its commitment to support the stock price (the purchase of the shares may drive up the share price) while profiting by disparity between the price at which it sold the shares and the lower price at which it repurchased them.⁸

⁸ When the bank is particularly worried that the share price will drop, it may sell even more than 15% of shares that the “Green Shoe” option allows. Essentially, the bank has then constructed a “naked short” position: it must buy back the excess shares, whether the share prices rise or drop. If the share price falls, it will once again have supported the price more effectively while profiting from its trading strategy. If the share price rises, however, it will need to purchase the additional shares at a loss.

The relationship between the underwriter and the portfolio company does not end in the weeks after the offering. Rather, at least in the United States, a complex relationship continues, with many points of interaction. These include the analyst coverage noted above,⁹ but also a variety of other roles. In virtually all cases, a U.S. investment bank will serve as a market maker: a trader responsible for insuring orderly day-to-day transactions in a security (including holding excess shares if necessary). In fact, the lead underwriter is virtually always the most important source of market-making activities in the months after the IPO. Finally, the underwriter of the IPO continues to serve as a financial advisor in most cases: about two-thirds of the firms completing a follow-on offering in the United States in the three years after the IPO employ the same underwriter.

⁹ Perhaps not surprisingly, it has been shown that investment banks issue more buy recommendations on companies that they underwrite than on other firms and that these recommendations seem to be excessively favorable (relative to the firms' subsequent performance).

Additional Information Sources

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